

Business Requirements Document (BRD) - HR Agentic Solution (MVP 1)

Executive Summary

The HR Agentic Solution is a secure, AI-driven virtual assistant designed to provide employees with immediate, conversational access to HR services. By seamlessly orchestrating workflows across core enterprise systems (WorkWeek, ServiceImmediately) and querying internal HR policies, the assistant will automate routine inquiries and facilitate self-service transactions.

To ensure strict enterprise governance and zero-trust security, this solution will be deployed on a secure, scalable agent orchestration platform, enforce traceably bounded execution, authenticate the automation's origin for all backend operations, and dynamically validate all interactions to protect against malicious instruction overrides and safeguard sensitive information.

1. Project Objectives

- **Deflect Tier 1 HR Inquiries:** Reduce routine HR and IT helpdesk ticket volume by at least 40% within the first six months by automating responses to common policy and status queries.
- **Streamline HR Transactions:** Enable employees to perform core self-service actions (leave submission, Incident ticket updates) conversationally without navigating complex backend UIs.
- **Validate Cross-System Orchestration:** Demonstrate the capability to chain actions across HR policies, WorkWeek, and ServiceImmediately to fulfill complex user intents, serving as a critical evaluation benchmark for prototype success before scaling.
- **Enhance Employee Experience:** Provide a unified conversational capability capable of orchestrating complex workflows across multiple backend systems (HCM, ITSM) and knowledge bases.
- **Ensure Enterprise AI Governance:** Maintain 100% visibility over the solution's deployment state, versioning, and authorized tool access limits.

- **Mitigate AI Risks:** Achieve zero policy violations and data leaks by intercepting and validating all user inputs and AI-generated outputs through dynamic interaction safeguards.
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2. Project Scope (MVP 1)

2.1. Functional Scope (In-Scope for MVP 1)

The solution will support the following conversational interaction types and capabilities:

- **Conversational User Interface (UI):** A standard, functional web-based chat interface or integration with a readily available enterprise chat client to facilitate direct user interaction and testing.
- **Policy & Informational Queries (RAG):** Answering questions related to HR policies, benefits, and procedures based strictly on approved static documents.
 - Question Types: "What is the policy on [Topic]?", "Am I eligible for [Benefit]?", "What is the process for [Action]?"
- **Employee Self-Service Transactions (WorkWeek):**
 - Read Actions: Fetching real-time employee profile details and leave balances.
 - Write Actions: Modifying specific personal contact information; submitting new leave requests.
- **Support Desk Management (ServiceImmediately):**
 - Read Actions: Checking status, priority, and comment history of specific tickets.
 - Write Actions: Creating new incidents, adding comments to existing tickets, and updating ticket status (e.g., to 'Resolved').
- **Cross-System Orchestration:** Chaining actions across domains (e.g., verifying policy eligibility via RAG, checking balance in WorkWeek, and creating a request in ServiceImmediately).

2.2. Data Scope (In-Scope for MVP 1)

The solution will operate upon and have access to the following specific data domains:

- **Static HR Policies:** A curated set of approved documents (PDF/Text) covering core HR domains (e.g., Leave Policies, Expense Guidelines, Remote Work Policy, Code of Conduct).
- **WorkWeek Data Elements:**
 - *Employee Profile:* Employee ID, Name, Email, Department, Role, Manager, Hire Date.
 - *Contact Info:* Personal Address, Personal Phone Number.
 - *Leave Data:* Accrued, Used, and Remaining balances for "Vacation" and "Sick" categories.
- **ServiceImmediately Data Elements:**
 - *Incident Record:* Ticket ID, Short Description, Detailed Description, Category, Priority level, State/Status, Assignee, Comments/Notes timeline.

2.3. Out of Scope for MVP 1

- Integration with systems other than WorkWeek, ServiceImmediately, and the designated Policy Repository.
 - Multi-Lingual capabilities
 - Processing of payroll data, performance reviews, or compensation details.
 - Voice-based interactions.
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3. MVP 1 Use Cases and Interaction Examples

To illustrate the solution's capabilities, the following table details the key MVP 1 Use Cases across single-domain inquiries and cross-system orchestration:

Use Case ID	Category	Triggering User Prompt (Example)	Systems Involved	System Actions / Expected Behavior
UC-1.1	Policy Q&A	"What is the company's bereavement	Policy Docs (RAG)	Query policy documents, retrieve

Use Case ID	Category	Triggering User Prompt (Example)	Systems Involved	System Actions / Expected Behavior
		<i>leave policy?"</i> or <i>"Are employees allowed to expense noise-canceling headphones?"</i>		relevant sections, and present a grounded answer with citation metadata (URLs/Deep Links).
UC-1.2	HR Self-Service	<i>"How many hours of PTO do I currently have accrued?"</i> or <i>"Please submit a time-off request for this coming Thursday and Friday."</i>	WorkWeek (HCM)	Retrieve current leave balances or validate and submit a time-off request in WorkWeek.
UC-1.3	IT Incident Mgmt	<i>"What is the status of ticket INC123456?"</i> or <i>"Create an IT ticket because my VPN connection keeps dropping."</i>	ServiceImmediately (ITSM)	Query status/comments or create a new support incident ticket in ServiceImmediately.

Use Case ID	Category	Triggering User Prompt (Example)	Systems Involved	System Actions / Expected Behavior
UC-2.1	Cross-System: Equipment Procurement	<i>"I just read the remote work policy and saw I'm eligible for a home office monitor. Can you verify my remote status and order one for me?"</i>	Policy Docs (RAG), WorkWeek, ServiceImmediately	1. Query remote work policy. 2. Verify user's location/status in WorkWeek. 3. Generate a hardware request in ServiceImmediately with shipping details.
UC-2.2	Cross-System: Medical Leave	<i>"I need to take short-term medical leave starting next Monday. What is the process, and can you set it up for me?"</i>	Policy Docs (RAG), WorkWeek, ServiceImmediately	1. Quote leave procedure from policy documents. 2. Submit Leave of Absence in WorkWeek. 3. Open ticket in ServiceImmediately to route user email access to manager.
UC-2.3	Cross-System: Relocation	<i>"I'm transferring to</i>	Policy Docs (RAG),	1. Quote relocation limits

Use Case ID	Category	Triggering User Prompt (Example)	Systems Involved	System Actions / Expected Behavior
		<i>the London office next month. Can you tell me the relocation allowance, update my record, and get my building access sorted?"</i>	WorkWeek, ServiceImmediately	from policy documents. 2. Prompt address update in WorkWeek. 3. Open facilities badge ticket in ServiceImmediately.

4. Functional Requirements

4.1. AI Governance & Security Infrastructure

Requirement ID	Requirement Name	Description
FR-1.1	Capability & Lifecycle Governance	The system must track ownership, version history, and enforce strict boundaries on authorized external tools (e.g., WorkWeek, ServiceImmediately, Document Repo). Any tool invocation or requests beyond these defined

Requirement ID	Requirement Name	Description
		capabilities must be blocked.
FR-1.2	Verification of Request Origin	All downstream calls must verify that they originate from an authorized automation entity acting on behalf of a specific user. Audit records must clearly differentiate between automated actions performed by the system and manual end-user inputs.
FR-1.3	Verification of Conversation Safety	All user inputs and generated outputs must be audited and verified before execution or display. Input Validation: The system must intercept and block prompt injection, jailbreak attempts, and off-topic interactions. Output Validation: The system must intercept response payloads to prevent toxic language, inaccurate/hallucinated data, or unauthorized extraction of sensitive info.
FR-1.4	Data Masking/Redaction	The system must detect and redact SPII (Sensitive

Requirement ID	Requirement Name	Description
		Personally Identifiable Information) from log files and conversational history to ensure privacy compliance.
FR-1.5	RBAC and Data Isolation	The system must enforce strict Role-Based Access Control (RBAC) to ensure that users can only access their own data or authorized information, preventing unauthorized cross-user data access.

4.2. Core Capabilities

Requirement ID	Requirement Name	Description
FR-2.1	Natural Language Understanding	The system must accurately parse user intent, accommodating typos, synonyms, and conversational context, subject to safety and input validation checks.
FR-2.2	Multi-Turn Dialog	The system must maintain state across a conversation while ensuring session memory does not cache sensitive data across different user sessions.

4.3. WorkWeek Integration (HCM)

Requirement ID	Requirement Name	Description & Operations
FR-3.1	Delegated Authorization	API calls to WorkWeek must pass a composite authentication token that scopes data retrieval strictly to the specific employee querying the system.
FR-3.2	Core Actions	The system must support the following operations: - Retrieve Employee Profile: Retrieve core work and contact metadata including employee name, email, department, role, manager, hire date, home address, and phone number. - Update Contact Information: Update the employee's personal home address and phone number in the system record. - Query Time-Off Balances: Check accrued, used, and remaining balances for both "Vacation" and "Sick" leave categories. - Submit Leave Request: Request time off specifying

Requirement ID	Requirement Name	Description & Operations
		start date, end date, leave type (Vacation or Sick), and work days, validating that the request does not exceed remaining accrued days.
FR-3.3	WorkWeek Operation Guardrails	The system must enforce validation logic on self-service transactions: - Balance Constraints: Prevent leave requests that exceed the caller's accrued vacation/sick balance. - Temporal Validity: Enforce chronological consistency for time-off dates (e.g. preventing past dates or start dates after end dates). - Format Restrictions: Validate input syntax for profile updates (e.g. phone numbers and email formats).
FR-3.4	Real-time Data Fetch	The system must fetch Employee Profile and PTO balances directly from WorkWeek on every query. No employee-specific dynamic data should be cached in the AI orchestration layer.

4.4. ServiceImmediately Integration (ITSM/HRSD)

Requirement ID	Requirement Name	Description & Operations
FR-4.1	Auditable Ticket Creation	When creating a ServiceImmediately ticket, the system logs must prevent audit ambiguity by explicitly recording the verified automation source that generated the request.
FR-4.2	Status Tracking and Ticket Management	The system must support the following operations: - Query Ticket Details: Retrieve current status, category, short description, priority, assignee, and the complete comment timeline for a specific incident ticket ID. - Create Incident Ticket: Open a new support ticket specifying requestor employee ID, category, short description, and priority level ('1 - Critical', '2 - High', '3 - Moderate', '4 - Low'). - Post Ticket Comment: Append update notes/comments to the ticket activity timeline. - Update Ticket Status: Transition the lifecycle state of a ticket (e.g. to 'Resolved' or 'Closed'),

Requirement ID	Requirement Name	Description & Operations
		including optional resolution notes.
FR-4.3	ServiceImmediately Operation Guardrails	The system must govern incident lifecycle transactions to prevent spam and audit corruption: - Transition Constraints: Validate that ticket status updates follow a logical lifecycle path (e.g. preventing direct transition from New to Closed). - Duplication Mitigation: Scan for potential duplicate tickets submitted in quick succession before creating new entries. - Priority Verification: Enforce alignment between ticket priority and incident descriptions (e.g., verifying Critical tags against defined criteria).

4.5. Policy Document Q&A (RAG System)

Requirement ID	Requirement Name	Description
FR-5.1	Document Ingestion	The system must connect to a centralized repository to ingest, chunk, and index HR policies.

Requirement ID	Requirement Name	Description
FR-5.2	Grounded Answers	The AI must only generate answers derived from the ingested policies. If the answer is not present, it must explicitly state so.
FR-5.3	Source Citation	Every policy-related answer must return metadata, URLs, or Deep Links rendered as a clickable citation to the exact document and section used.
FR-5.4	Policy Retrieval Guardrails	The system must apply constraints to conversation scope and generation safety: - Strict Grounding: Refuse to answer questions when retrieved document context is insufficient, preventing hallucinations. - Domain Containment: Enforce topic boundaries, rejecting prompts that fall outside corporate HR policy (e.g. general coding questions or personal queries). - Citation Integrity: Ensure all generated citations resolve to active, verified policy documents.

Requirement ID	Requirement Name	Description
FR-5.5	Document Sync Latency	The system must reflect updates to policy documents in the Knowledge Base within [X] hours/minutes of the document being updated in the source repository.

5. Non-Functional Requirements

5.1. Security and Privacy

Requirement ID	Requirement Name	Description
NFR-1.1	Safety for AI Interactions	The system must ensure that all AI-driven interactions remain safe and aligned. The model must never generate unsafe, toxic, or harmful responses, and it must recognize, block, and refuse to process unsafe or malicious user requests.
NFR-1.2	Audit Logging	Every action taken by the system, including denied actions blocked by input/output validation checks, must be logged.
NFR-1.3	Compliance Adherence	All data processing and storage must comply with relevant data privacy

Requirement ID	Requirement Name	Description
		regulations (e.g., GDPR, local labor laws).

5.2. Performance and Scalability

Requirement ID	Requirement Name	Description
NFR-2.1	Latency	The system must begin generating a response within 10 seconds. The addition of input/output safety scanning must not add more than 300ms of latency per turn.
NFR-2.2	Availability	The solution must guarantee 99.9% uptime, aligning with standard enterprise SaaS SLAs.
NFR-2.3	Asynchronous Processing	Long-running operations or parallel backend calls must be handled asynchronously to prevent blocking the main conversational flow.

5.3. Quality & Accuracy

Requirement ID	Requirement Name	Description
NFR-3.1	Accuracy Rate	The Agentic system must achieve an accuracy rate of >95% on a predefined benchmark set of policy questions, with 0% hallucinated policies.

5.4. Resilience & Error Handling

Requirement ID	Requirement Name	Description
NFR-4.1	Graceful Failure Handling	If an integrated system (WorkWeek, ServiceImmediately, or Policy Repo) is unavailable or returns an error, the Agent must notify the user with a clear, non-technical message (e.g., "Service is temporarily unavailable") and avoid exposing stack traces or internal error codes.
NFR-4.2	Transient Fault Tolerance	The system must implement retry logic (e.g., exponential backoff) for transient failures (e.g., network timeouts, rate limits) before reporting a service failure to the user.
NFR-4.3	Orchestration Consistency	For Cross-System Orchestration (UC-2.x), if one step fails (e.g., Leave submitted in WorkWeek but Ticket creation fails in ServiceImmediately), the system must log the failure clearly and provide instructions for manual follow-up or attempt compensating actions to ensure data consistency.

6. MVP 1 Implementation Constraints

To scope the initial release effectively, the following boundaries and constraints apply to MVP 1:

- **Authentication & Credentials:** The system will use functional test credentials for backend integrations. Integrations with enterprise identity management systems (such as Active Directory, Okta, or Single Sign-On / SSO) are excluded from this release.

- **Tenancy Scope:** The initial implementation targets a single-tenant environment. Multi-tenancy support is not required or supported for MVP 1.

7. Success and Evaluation Criteria

To evaluate the MVP 1 prototype for potential large-scale rollout, the business and HR teams will assess the solution against the following criteria:

Evaluation Category	Success Metric / Criterion	Target / Benchmark
Policy Q&A Accuracy	Precision and recall of answers derived from policy documents (RAG).	>= 95% Accuracy on benchmark questions; 0% Hallucination of policy facts.
Transaction Integrity	Successful execution of self-service actions (leave submission, ticket creation) in backend systems.	100% Transaction Correctness (no data corruption or unauthorized updates).
Cross-System Orchestration	Ability to chain actions across Policies, WorkWeek, and ServiceImmediately.	Pass/Fail on all defined Cross-System Use Cases (UC-2.x).
Safety & Guardrail Efficacy	Ability of the system to identify and block malicious, unsafe, or off-topic prompts.	100% Detection of known prompt injection/jailbreak test cases; < 1% False Positives (blocking legitimate queries).
Response Latency	Time taken by the system to begin generating a response.	< 10.0 Seconds average response time; Safety scanning overhead < 300ms.

Auditability & Traceability	Verification that all actions (allowed and blocked) are logged with clear origin indicators.	100% Log Coverage for all API interactions and safety blocks.
Resilience & Error Handling	System behavior during simulated downtime of integrated systems or unavailability of requested information	100% Graceful degradation; No technical leaks; Clear fallback instructions provided to user.
User Experience (NLU)	Robustness of intent detection against typos, synonyms, and conversational context.	Qualitative "Pass" on ease of use and natural flow during evaluator testing.